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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fourth Semester B.E. (CSE) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

20CS405 - MICROPROCESSORS AND PERIPHERALS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **One full** question from **each Unit**.*

		Marks	BT*	CO*	PO*
Unit – I					
1.	a) Draw and explain the programming model of the 8086 microprocessor.	7	L*2	1	1
	b) Explain clearly with example how physical address is generated.	6	L2	1	2
	c) Develop an assembly language program to reverse a string in the memory and check whether it is a palindrome or not.	7	L3	1	3
2.	a) List all addressing modes available in Intel 8086 microprocessor and explain any four with examples.	8	L2	1	1
	b) Explain the following instructions with examples. i) XCHG ii) DIV iii) DAA	6	L2	1	2
	c) Develop an assembly language program to search an 8 bit key value in an 8 bit array of values. Display the key position if the key is found otherwise display message key not found. Assume that the key and array elements are stored in the memory.	6	L3	1	3
Unit – II					
3.	a) Write a macro to clear the screen and set the cursor at the specified location on the screen.	6	L3	2	2
	b) Explain the following assemble directives with examples. i) EQU ii) SEGMENT iii) ORG iv) DW	8	L2	2	2
	c) What is a macro? Differentiate between Macro and procedure.	6	L1	2	1
4.	a) Write assembly code to perform the following operations: i) Set system date ii) Write to a file iii) Open a file iv) Buffered key board input.	8	L3	2	3
	b) Write an ALP code to covert two ASCII codes into packed BCD number.	6	L3	2	3
	c) Explain conditional & unconditional jump instructions with examples.	6	L1	2	2
Unit – III					
5.	a) What is Microcontroller? Differentiate between CISC & RISC.	6	L2	3	1
	b) List the different modes of ARM. Also explain the complete ARM register set.	8	L2	3	2
	c) With neat diagram explain CPSR.	6	L2	3	2
6.	a) With a neat diagram explain the arm core dataflow model.	10	L2	3	1
	b) What is pipeline in ARM? Illustrate with an example. Show the pipeline stages of ARM7, ARM9 & ARM10.	10	L2	3	2
Unit – IV					
7.	a) Explain the analog and digital I/O built in functions in embedded C.	10	L2	4	2
	b) Explain the serial monitoring functions. Give examples.	10	L3	4	2
8.	a) Explain with program how to interface keypad with microcontroller.	10	L3	4	3

P.T.O.

- b) Explain with program how stepper motor can be interfaced with microcontroller.

10 L3 4 3

Unit – V

9. a) Explain the following functions of 8086 microprocessor.
i) READY ii) ALE iii) NMI iv) HLDA v) DEN
- b) Explain Interrupt Vector Table along with the action taken by 8086 MP when an interrupt occurs.
10. a) Briefly explain the operation of 8259, with a neat block diagram and initialization command words.
- b) State and explain instruction format for MOV instruction to transfer data between register and memory. Also generate opcode for the following.
i) MOV AX[BX] ii) MOV CX.[1000H].

10 L2 5 1

10 L2 5 2

10 L3 5 2

10 L3 5 3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
